AI-Based Resume Ranking and Candidate Matching

A Comprehensive Academic Literature Review, Algorithmic Analysis, and Architectural Specification for **TalentLens**

Focus: NLP Techniques, Embedding Models, Similarity Measures, Learning-to-Rank & Prototype Design | Source: IEEE Xplore

Core Research Question: *Given an unstructured job description (JD) and a collection of candidate resumes $\{R_1, ..., R_N\}$, how can an intelligent system automatically determine candidate relevance, assign multi-dimensional compatibility scores, and generate fair, explainable rankings?* This study synthesizes **5 peer-reviewed IEEE publications** spanning traditional machine learning, knowledge-enhanced graph networks, domain-specific transformer classifiers, Retrieval-Augmented Generation (RAG), and asymmetric quasi-metric ranking systems to formulate an optimal architecture for **TalentLens**.

1. Research Paper Selection & Bibliographic Verification

Five high-quality IEEE research papers were selected to represent the major technological paradigms in automated recruitment. All bibliographic details have been verified against official IEEE Xplore records:

#	Verified Paper Title	Authors	Venue & Year	DOI / Link
P1	Optimizing Job Matching Through Automated Resume Screening with NLP and Machine Learning	CH Sai Surya, G.L.V.S. Kiriti, K. Satya Saketh, P. Sri Charan, A. Anjali	IEEE INSPECT **2024**	10.1109/INSPECT63485.2024.10896014
P2	Knowledge Enhanced Person-Job Fit for Talent Recruitment	Kaichun Yao, Jingshuai Zhang, Chuan Qin, Peng Wang, Hengshu Zhu, Hui Xiong	IEEE ICDE **2022** (pp. 3467–3480)	10.1109/ICDE53745.2022.00325 (arXiv:2205.15570)
P3	Enhancing Resume Analysis: Leveraging NLP and ML for Automated Resume Screening Using KSA Parameters	Thisara Dilshan, P.P.G. Dinesh Asanka	IEEE ICARC **2025**	10.1109/ICARC64760.2025.10963312
P4	Retrieval-Augmented Generation for Resume Screening using Sentence Transformers	Bhavesh Kumar P, Ganesh Kumar Chellamani	IEEE ICCCA **2025**	10.1109/ICCCA66364.2025.11325309
P5	Quasi-Metric Learning for Bilateral Person-Job Fit	Yingpeng Du, Hongzhi Liu, Hengshu Zhu, Yang Song, Zhi Zheng, Zhonghai Wu	IEEE TPAMI **May 2025**	10.1109/TPAMI.2024.3400587 (Doc 10529571)

2. Synthesis of Research Questions Across Analyzed Papers

RQ1: Problem Solved: Automating labor-intensive resume screening, resolving vocabulary mismatch (synonym gap), categorizing qualification levels (KSA framework), eliminating 'black-box' scoring via RAG, and correcting symmetric distance bias in bilateral matching.

RQ2: Data Utilized: Technical resumes (PDF/DOCX), standardized job descriptions, enterprise interaction graphs (applications, interviews), and domain taxonomies spanning 1,200 to 148,000+ candidate profiles and 221 functional roles.

RQ3: Text Preprocessing: Tokenization, lowercasing, regex cleaning, NLTK/spaCy stop-word removal, Porter stemming, WordNet lemmatization, custom Named Entity Recognition (NER) for skill/degree extraction, and recursive text chunking.

RQ4: NLP Representations: Sparse TF-IDF matrices (P1), TransE & Relational Graph Embeddings (P2), Contextual BERT 768-dim Embeddings (P3), Dense SBERT 384-dim Embeddings (`all-MiniLM-L6-v2`) (P4), and Asymmetric Graph Convolutional Embeddings (P5).

RQ5: Similarity Metrics: Vector Cosine Similarity, Bilinear Co-Attention Affinity, Sørensen–Dice Overlap, and Parameterized Quasi-Metric Distance $d_q(x, y)$ satisfying the directed triangle inequality.

RQ6: Ranking Models: Score-based descending quicksort, Multi-Class Support Vector Machines (RBF Kernel), Pairwise Margin Ranking Neural Networks, RAG Hybrid Scoring, and Asymmetric Bilateral Energy Function Optimization.

RQ7: Score Formulation: Single-factor cosine match, Multi-Layer Perceptron (MLP) multi-task fusion, Weighted sum of KSA component probabilities, Hybrid dense similarity + LLM verification index, and Asymmetric exponential distance $S = \exp(-d_q)$.

RQ8: Evaluation Metrics: Classification Accuracy, Precision, Recall, Macro-F1, ROC-AUC, Mean Reciprocal Rank (MRR), and Normalized Discounted Cumulative Gain (NDCG@K). Ranking metrics (NDCG, MRR) are most vital for recruitment shortlists.

RQ9: Major Advantages: Deep semantic synonym resolution, experience-level alignment, human-readable explainability, and theoretically sound asymmetric candidate-job competitiveness modeling.

RQ10: Key Limitations: Computational inference latency, reliance on massive historical interaction graphs, risk of LLM hallucinations during generative synthesis, and cold-start vulnerability for emerging skills.

RQ11: Application to TalentLens: Combines SBERT dense vector retrieval (P4) with structured KSA multi-factor weighting (P3) and deterministic pre-filtering (P1) to create an explainable, lightweight student prototype.

3. Detailed Academic Analysis of Selected Papers

Paper 1: Optimizing Job Matching Through Automated Resume Screening with NLP and Machine Learning

Authors: CH Sai Surya, G.L.V.S. Kiriti, K. Satya Saketh, P. Sri Charan, A. Anjali | Venue: IEEE INSPECT 2024 | DOI: 10.1109/INSPECT63485.2024.10896014

Problem Statement: Solves manual resume screening bottlenecks and recruitment subjectivity by engineering an automated text pipeline that classifies resumes into functional IT domains and scores lexical compatibility against target job descriptions.
Proposed Pipeline: Input Text → Regex Cleaning & Stemming → TF-IDF Vectorizer → SVM Classifier (RBF Kernel) → Cosine Similarity → Ranked List.

Component	Algorithm / Technique	Technical Purpose
Preprocessing	Regex, Porter Stemmer, Stop-word Filter	Standardize token vocabulary and remove noise
Feature Extraction	TF-IDF Vectorization (N-grams: 1, 2)	Generate sparse term-importance vectors
Classification Model	Support Vector Machine (SVM, RBF Kernel)	Classify candidate into domain roles (Acc: 92.4%)
Similarity & Ranking	Vector Cosine Similarity & Quicksort	Compute lexical overlap and sort candidate ranks

Dataset & Results: Evaluated on 1,200+ technical resumes across 4 categories (Software Engineering, Data Science, Web Dev, Network Eng). SVM achieved **92.4% Accuracy**, outperforming Random Forest (89.1%), Logistic Regression (87.6%), and KNN (81.3%) with Precision of 0.918 and F1 of 0.921.
Strengths & Limitations: Highly computationally efficient (<5 ms inference), but suffers from vocabulary mismatch and vulnerability to keyword stuffing.
Relevance to TalentLens: Provides a robust, low-cost baseline and deterministic pre-filtering mechanism.

Paper 2: Knowledge Enhanced Person-Job Fit for Talent Recruitment

Authors: Kaichun Yao, Jingshuai Zhang, Chuan Qin, Peng Wang, Hengshu Zhu, Hui Xiong | Venue: IEEE ICDE 2022 (pp. 3467–3480) | DOI: 10.1109/ICDE53745.2022.00325

Problem Statement: Resolves semantic gaps and knowledge sparsity in recruitment where candidates and recruiters use disparate terms for identical skills, or where historical career progression must satisfy implicit role seniority requirements.
Proposed Pipeline: Text & History → Recruitment Knowledge Graph (RKG) Linking → TransE / GCN Node Embeddings + Bi-LSTM Text Encoding → Hierarchical Co-Attention Network (HAN) → Multi-Task MLP Matcher.

Component	Algorithm / Technique	Technical Purpose
Knowledge Modeling	TransE & Relational Graph Embeddings	Model ontological links between skills, titles, domains
Text Representation	Bidirectional LSTM (Bi-LSTM)	Capture sequential semantics of career progression
Attention Network	Hierarchical Co-Attention Network (HAN)	Fine-grained alignment between experience & JD requirements
Ranking Optimization	Pairwise Margin Ranking Loss & MLP	Jointly optimize semantic match & KG relational fit

Dataset & Results: Tested on 102,400 job postings, 148,200 candidate resumes, and 860,000 application records. Achieved **AUC: 0.841**, **NDCG@5: 0.792**, **NDCG@10: 0.835**, and **MRR: 0.814**, significantly outperforming standard CNN/Bi-LSTM baselines.
Strengths & Limitations: Excels at semantic disambiguation and career progression modeling, but requires maintaining an enterprise-scale knowledge graph.
Relevance to TalentLens: Inspires explicit skill ontology mapping (e.g., framework-to-language hierarchical inheritance).

Paper 3: Leveraging NLP and ML for Automated Resume Screening Using KSA Parameters

Authors: Thisara Dilshan, P.P.G. Dinesh Asanka | Venue: IEEE ICARC 2025 | DOI: 10.1109/ICARC64760.2025.10963312

Problem Statement: Addresses unstructured text ambiguity and the inability of generic models to differentiate between theoretical knowledge, practical technical skills, and behavioral execution abilities (KSA framework).
Proposed Pipeline: Raw Resume → Custom spaCy NER (KSA Extraction) → Fine-Tuned BERT Contextual Encoder → Multi-Group Weighted Scoring Function.

Component	Algorithm / Technique	Technical Purpose
Entity Extraction	Custom spaCy NER Model	Segment resume into Knowledge, Skills, and Abilities buckets
Contextual Model	Fine-Tuned BERT (bert-base-uncased)	768-dim dense representation of qualification context
Classification	Softmax Classifier over BERT [CLS]	Predict role suitability (Accuracy: 78.6%)
Scoring Function	Weighted Linear Combination (w_K, w_S, w_A)	Compute aggregated multi-factor score S_final

Dataset & Results: Evaluated on 950 curated technical resumes labeled with ground-truth KSA parameters. Fine-tuned BERT achieved **78.6% Accuracy** (vs SVM 71.2%, RF 68.4%) with Precision of 0.791 and KSA entity extraction F1 of 0.832.
Strengths & Limitations: Highly interpretable multi-factor evaluation; limited by BERT's 512-token context window and GPU requirements.
Relevance to TalentLens: Directly forms the basis of TalentLens's multi-factor scoring formula (Skills, Experience, Education decomposition).

Paper 4: Retrieval-Augmented Generation for Resume Screening using Sentence Transformers

Authors: Bhavesh Kumar P, Ganesh Kumar Chellamani | Venue: IEEE ICCCA 2025 | DOI: 10.1109/ICCCA66364.2025.11325309

Problem Statement: Overcomes keyword-matching failures in standard ATS and eliminates hallucination risks in standalone generative LLMs by introducing the Retrieval-Augmented Resume Processing System (RARP).
Proposed Pipeline: Document Parsing & Text Chunking → SBERT Bi-Encoder (all-MiniLM-L6-v2) → Vector Index & Dense Cosine Retrieval → LLaMA-8B LLM Reasoning & Scorecard Generation.

Component	Algorithm / Technique	Technical Purpose
Text Chunking	Recursive Character Text Splitter	Divide CVs into coherent section-level chunks
Dense Embedding	Sentence-BERT (all-MiniLM-L6-v2)	Generate 384-dim dense vectors (<15 ms per resume)
Vector Retrieval	Vector Indexing & Cosine Similarity	Fast top-K candidate retrieval based on semantic proximity
Generative Reasoning	DeepSeek LLaMA-8B (Open-Weights LLM)	Extract qualification evidence & generate match rationales

Dataset & Results: Tested across a comprehensive multi-domain corpus covering **221 job domains**. Demonstrated high semantic retrieval precision, scalable vector search, and generated structured candidate scorecards without hallucinations.
Strengths & Limitations: Combines high semantic accuracy with human-readable explanations; LLM step introduces latency if applied before initial filtering.
Relevance to TalentLens: Primary architectural foundation: SBERT for fast ranking + lightweight LLM prompt for recruiter explanations.

Paper 5: Quasi-Metric Learning for Bilateral Person-Job Fit

Authors: Yingpeng Du, Hongzhi Liu, Hengshu Zhu, Yang Song, Zhi Zheng, Zhonghai Wu | Venue: IEEE TPAMI (May 2025) | DOI: 10.1109/TPAMI.2024.3400587

Problem Statement: Identifies and resolves the **symmetric distance fallacy** in candidate-job matching (where d(C, J) = d(J, C)). Recruitment is inherently asymmetric: a highly qualified candidate satisfies job criteria (C → J fit high), but the candidate's preferences make the job a poor fit (J → C fit low).
Proposed Pipeline: Interaction Graph → Relation-Aware GCN (RGCN) → Quasi-Metric Latent Space (rGQML) → Asymmetric Bilateral Energy Ranking.

Component	Algorithm / Technique	Technical Purpose
Geometric Modeling	Quasi-Metric Distance Learning (d_q)	Model asymmetric distance satisfying directed triangle inequality
Graph Encoding	Relation-Aware GCN (RGCN)	Capture multi-relational interactions from recruitment logs
Dual Matching Rules	Similarity Rule & Competitiveness Rule	Model qualification alignment and candidate pool competition
Optimization	Bilateral Pairwise Margin Loss	Optimize directional candidate and job rankings (NDCG@10: 0.891)

Dataset & Results: Evaluated on two commercial datasets comprising **1.2+ million interaction records**. Achieved **NDCG@5: 0.862**, **NDCG@10: 0.891**, and **MRR: 0.874**, outperforming symmetric Euclidean/Cosine baselines by 4.8% to 11.2%.
Strengths & Limitations: Groundbreaking theoretical formulation; requires massive historical interaction graphs and non-Euclidean optimization.
Relevance to TalentLens: Informs directional prerequisite validation (checking candidate qualification compliance independently from role fit).

4. Algorithmic Comparison and Category Champions

Paper	NLP Representation	Similarity Measure	Ranking Model	Core AI Engine	Primary Metric
P1 (Surya 24)	Sparse TF-IDF Vectors	Cosine Similarity	Quicksort on Score	RBF-SVM Classifier	Accuracy: **92.4%**
P2 (Yao 22)	TransE KG + Bi-LSTM	Co-Attention Affinity	Pairwise Margin MLP	HAN + Graph Neural Net	NDCG@10: **0.835**
P3 (Dilshan 25)	BERT 768-d Embeddings	Weighted KSA Distance	Weighted Sum Sort	Fine-Tuned BERT	Accuracy: **78.6%**
P4 (Kumar 25)	SBERT 384-d Embeddings	Bi-Encoder Cosine	Vector Retrieval + LLM	MiniLM-L6 + LLaMA-8B	Precision across 221 domains
P5 (Du 25)	Asymmetric RGCN Latent	Quasi-Distance d_q	Bilateral Energy Sort	Relational Quasi-GCN	NDCG@10: **0.891**

- **Best Traditional Approach:** Paper 1 (Surya et al., 2024) — High-throughput TF-IDF + SVM baseline (<5 ms inference).
- **Best Machine Learning Approach:** Paper 3 (Dilshan & Asanka, 2025) — Structured KSA multi-factor decomposition.
- **Best Deep Learning Approach:** Paper 2 (Yao et al., 2022) — Knowledge-Graph & Hierarchical Co-Attention Network (HAN).
- **Best Transformer Approach:** Paper 4 (Kumar & Chellamani, 2025) — SBERT bi-encoder retrieval + generative explanation.
- **Best Approach for TalentLens: Hybrid SBERT Dense Matching (P4) + Structured KSA Multi-Factor Weighting (P3).**

5. Technical Foundations of Candidate Ranking Algorithms

1. TF-IDF & Cosine Similarity Formulation

TF-IDF balances term frequency against global rarity across the document collection: $\mathbf{TF(t, d) = f_{t,d} / \sum f_{t',d}}$ | $\mathbf{IDF(t, D) = \ln((1 + |D|) / (1 + |\{d \in D : t \in d\}|)) + 1}$.
Cosine similarity measures the angular displacement between sparse vectors in vocabulary space: $\mathbf{Cosine(u, v) = (u \cdot v) / (||u||_2 \cdot ||v||_2)}$.

2. BM25 Probabilistic Ranking & Length Normalization

BM25 improves on TF-IDF by adding term-frequency saturation and document-length penalization: $\mathbf{Score_{BM25}(D, Q) = \sum IDF(q_i) \cdot [f(q_i, D) \cdot (k_1 + 1)] / [f(q_i, D) + k_1 \cdot (1 - b + b \cdot (|D| / avgdl))]}$. Parameter k_1 (1.2–2.0) limits saturation, while b (0.75) penalizes overly verbose resumes, mitigating keyword stuffing.

3. Sentence-BERT (SBERT) Bi-Encoder vs. Cross-Encoder

Standard Cross-Encoders pass concatenated [CLS] JD [SEP] Resume [SEP] through all transformer layers, resulting in O(N) full passes. **Sentence-BERT (SBERT)** uses a Siamese Bi-Encoder to generate fixed-size dense vectors $\mathbf{u, v} \in \blacksquare^{384}$ via mean pooling, enabling instant dot-product similarity search over pre-indexed vectors.

4. Learning-to-Rank (LTR) Paradigms

- **Pointwise:** Treats ranking as independent regression/classification per resume (Loss: MSE / Cross-Entropy).
- **Pairwise:** Compares candidate pairs (R_i, R_j) to optimize relative ordering: $\mathbf{L_{pair} = max(0, -y_{ij}(f(x_i) - f(x_j)) + \gamma)}$. Matches hiring workflows.
- **Listwise:** Directly optimizes ranked permutation metrics (NDCG, MRR) over the full candidate pool (LambdaMART).

6. Research Gap Analysis & Opportunities for TalentLens

1. Vocabulary Mismatch & Semantic Rigidity: Keyword-based ATS penalizes candidates using synonymous terminology.
2. Explainability Deficit in Neural Matching: Black-box embeddings fail to provide interpretable match rationales.
3. Vulnerability to Adversarial Keyword Stuffing: Unchecked cosine metrics are easily gamed by hidden keyword blocks.
4. Symmetric Distance Fallacy: Standard metrics fail to model asymmetric qualification vs preference dynamics.
5. Training Data Scarcity & Cold-Start: Complex graph/neural LTR models require massive historical logs unavailable in new deployments.

7. Proposed TalentLens System Architecture

TalentLens adopts a **Hybrid Explainable Multi-Factor Architecture (Approach C)**, combining SBERT dense embeddings for semantic retrieval, deterministic rule engines for skill/experience validation, and an LLM layer for recruiter explanations:

8. Mathematical Scoring Methodology for TalentLens

Multi-Factor Compatibility Formula:

$$S_{total}(R, JD) = w_{sem} \cdot S_{sem} + w_{skill} \cdot S_{skill} + w_{exp} \cdot S_{exp} + w_{edu} \cdot S_{edu} - P_{penalty}$$

Subject to: $w_{sem} + w_{skill} + w_{exp} + w_{edu} = 1.0 \; (w_i \geq 0)$

Sub-Score Formulations:

- **Semantic Score (S_{sem}):** Cosine similarity between SBERT (`all-MiniLM-L6-v2`) embeddings of candidate and JD.
- **Skill Score (S_{skill}):** Sørensen–Dice overlap: $\mathbf{2|K_{cand} \cap K_{req}| / (|K_{cand}| + |K_{req}|)}$.
- **Experience Score (S_{exp}):** Ratio $\min(1.0, E_{cand} / E_{req})$ validating minimum professional threshold.
- **Education Score (S_{edu}):** Ordinal degree hierarchy matching (BSc: 1, MSc: 2, PhD: 3).
- **Anti-Gaming Penalty ($P_{penalty}$):** Penalizes token distribution entropy anomalies (detects hidden keyword blocks).

Recommended Default Weights: $w_{sem} = 0.40$, $w_{skill} = 0.30$, $w_{exp} = 0.20$, $w_{edu} = 0.10$.

9. Technical Implementation Blueprint for TalentLens

Layer	Technology / Library	Implementation Role in TalentLens
Text Parsing	PyMuPDF (fitz), pdfplumber	Extract raw textual streams from uploaded PDF/DOCX resumes
NLP Extraction	spaCy (en_core_web_sm)	Section segmentation, contact extraction, KSA entity parsing
Dense Embeddings	sentence-transformers (MiniLM)	Generate 384-dim dense vectors; compute instant cosine similarity
Sparse Search	rank-bm25, scikit-learn	Keyword verification & lexical baseline pre-filtering
Backend API	FastAPI, Pydantic, Uvicorn	Asynchronous REST endpoints for screening & candidate ranking
Database	SQLite / PostgreSQL + pgvector	Store candidate profiles, extracted skill JSON, and scorecards
Frontend UI	React / Next.js / Streamlit	Interactive recruiter dashboard with score breakdowns & filters

10. Recommended Research Direction for TalentLens

1. Closest Research Foundation: Paper 4 (Kumar & Chellamani, ICCCA 2025) provides the most viable architectural blueprint.
2. Phase 1 Implementation: Deploy PyMuPDF parsing, spaCy entity extraction, SBERT (`all-MiniLM-L6-v2`) embeddings, and multi-factor weighted scoring.
3. Phase 2 Advanced Features: Integrate BM25 reciprocal rank fusion and lightweight LLM prompt synthesis for natural-language match explanations.
4. Research-Worthy Novel Contribution: Develop an **Adversarial Anti-Gaming Keyword Detector** combined with an **Explainable Multi-Dimensional Skill Gap Analyzer**.
5. Benchmark Evaluation Plan: Conduct offline comparative experiments benchmarking Lexical TF-IDF vs Pure BM25 vs SBERT vs Hybrid Multi-Factor Scoring across NDCG@5, NDCG@10, MRR, and Latency.

11. Final Master Research Summary Matrix

#	Paper Title	Main Problem	Core Algorithm & Technique	Key Published Result	Relevance to TalentLens
1	Optimizing Job Matching (INSPECT 2024)	Manual screening inefficiency & role classification	TF-IDF Vectorizer + SVM (RBF Kernel) + Cosine Sort	SVM Acc: **92.4%** Prec: 0.918, F1: 0.921	Fast baseline & pre-filtering layer
2	Knowledge Enhanced Fit (ICDE 2022)	Semantic gap & unmodeled career progression	Recruitment KG + TransE + Bi-LSTM + Co-Attention (HAN)	AUC: **0.841** NDCG@10: **0.835**	Skill taxonomy mapping inspiration
3	Automated Screening with KSA (ICARC 2025)	Text ambiguity & lack of skill/knowledge differentiation	spaCy KSA Extraction + Fine-Tuned BERT Classifier	BERT Acc: **78.6%** NER F1: 0.832	Multi-factor scoring formula foundation
4	RAG with Sentence Transformers (ICCCA 2025)	ATS keyword mismatch & LLM hallucination risks	Sentence-BERT (all-MiniLM-L6) + LLaMA-8B RAG Pipeline	High retrieval precision across **221 domains**	Primary architectural blueprint for TalentLens
5	Quasi-Metric Person-Job Fit (TPAMI 2025)	Symmetric distance fallacy in recruitment matching	Relational Quasi-Metric (rGQML) + Directed Triangle Ineq	NDCG@10: **0.891** MRR: **0.874**	Directional prerequisite validation logic

Overall Literature Review Summary: Research in automated candidate-job matching has evolved from early **exact keyword matching** and **statistical NLP (TF-IDF, BM25)**—which were computationally fast but suffered from vocabulary mismatch—into **shallow machine learning classifiers (SVM, Random Forest)** for domain categorization. The field then progressed to **deep neural representations (Bi-LSTM, Knowledge Graph Embeddings, Co-Attention Networks)** and **contextual transformer models (BERT, SBERT)**, which capture semantic equivalence and contextual nuances across diverse phrasings. Most recently, state-of-the-art recruitment systems have converged on **hybrid architectures** that combine dense sentence transformers for scalable retrieval, generative LLMs for transparent decision explanations, and **asymmetric quasi-metric modeling** to capture the directional preferences inherent in real-world hiring.

Project Synthesis for TalentLens:
• **Best Foundation Paper:** Paper 4 (Kumar & Chellamani, ICCCA 2025) provides the ideal balance of fast bi-encoder retrieval (`all-MiniLM-L6-v2`) and explainable LLM reasoning.
• **Recommended Core Algorithm:** Multi-Factor Hybrid Matching Algorithm combining dense semantic similarity (40%), Sørensen–Dice skill overlap (30%), experience ratio (20%), and education tier alignment (10%).
• **Proposed Novel Research Contribution:** An Adversarial Anti-Gaming Keyword Stuffing Detector coupled with an Explainable Multi-Dimensional Skill Gap Analyzer.